

POINT PETRE, ON, Canada

WMO: 714300

Lat: 43.8396N

Lon: 77.1530W

Elev: 258

StdP: 14.56

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	2.9	-13.4	2.7	-1.9	-7.4	3.7	4.6	44.8	24.9	41.3	28.2	12.5	30	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	79.4	72.8	77.6	71.5	76.0	70.3	74.4	77.9	72.8	76.4	71.4	74.9	10.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	124.5	76.9	71.5	117.6	75.4	70.1	111.6	74.2	38.1	78.2	36.6	76.6	35.4	75.0	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.3	31.8	28.2	DB	-9.4	82.8	5.5	3.1	-13.3	85.0	-16.5	86.8	-19.6	88.5	-23.5	90.8	(4)
			WB	-10.0	75.9	5.4	2.3	-13.9	77.6	-17.1	78.9	-20.1	80.2	-24.0	81.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	23.3	25.1	31.6	41.0	51.3	60.7	68.3	68.9	62.5	51.3	40.6	30.8
	DBStd	17.45	11.42	9.06	8.08	5.72	5.68	5.37	4.50	4.60	6.34	7.16	7.64	9.30
	HDD50	3382	827	698	571	273	52	1	0	0	2	72	291	595
	HDD65	7117	1292	1118	1036	719	427	150	20	17	119	426	733	1060
	CDD50	2067	0	0	1	4	91	320	568	586	377	113	7	0
	CDD65	327	0	0	0	0	1	20	123	138	44	1	0	0
	CDH74	866	0	0	0	0	1	26	337	413	89	0	0	0
	CDH80	39	0	0	0	0	0	1	15	20	3	0	0	0
Wind	WSAvg	13.5	17.1	15.9	14.1	13.4	11.1	9.8	9.8	10.7	12.1	14.9	16.1	17.3
Precipitation	PrecAvg	35.2	2.6	2.2	2.5	3.1	3.1	3.1	3.4	3.1	3.1	3.3	2.8	2.8
	PrecMax	42.9	5.6	4.1	4.5	5.8	5.7	5.9	7.6	5.0	5.4	6.3	4.0	4.9
	PrecMin	27.8	1.4	0.8	0.9	0.9	1.1	0.7	1.3	1.2	1.5	1.1	1.0	1.5
	PrecStd	3.7	1.0	0.7	0.9	1.2	1.3	1.5	1.7	0.9	1.1	1.4	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	42.4	52.9	59.1	70.8	77.1	81.9	82.4	79.2	69.1	57.8	51.7
		MCWB	43.5	40.3	47.6	50.7	62.7	69.2	73.8	76.0	73.1	65.5	55.0	49.9
	2%	DB	42.2	40.1	46.6	54.1	66.2	73.7	79.4	79.7	76.5	66.5	55.3	48.3
		MCWB	40.4	38.0	42.7	47.4	60.0	67.6	72.5	73.3	71.0	62.8	52.3	46.5
	5%	DB	40.1	38.3	43.5	51.0	62.6	71.1	77.4	77.9	74.2	64.2	53.5	45.5
		MCWB	38.4	36.4	40.5	45.7	57.0	66.0	71.4	71.8	69.2	60.6	50.9	43.3
	10%	DB	37.8	36.6	41.1	48.6	59.4	68.8	75.7	76.3	72.1	62.1	51.4	42.9
		MCWB	36.1	34.7	38.5	44.2	54.8	64.2	70.0	70.6	67.7	58.7	48.5	40.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	40.7	49.6	53.2	65.1	71.7	76.1	77.3	74.8	66.8	56.0	50.4
		MCDB	44.9	41.8	52.6	56.3	69.5	75.1	79.3	80.4	77.9	68.3	57.4	51.0
	2%	WB	41.0	38.5	43.5	49.2	61.1	69.1	74.1	74.9	72.3	63.4	53.2	46.9
		MCDB	42.1	39.7	45.4	52.1	64.9	72.3	77.8	78.4	75.4	65.9	54.7	48.1
	5%	WB	38.7	36.7	40.9	46.7	58.2	66.9	72.4	73.2	70.3	61.3	51.1	43.5
		MCDB	39.9	38.0	43.1	50.2	61.7	70.2	76.3	76.7	73.3	63.7	53.1	45.4
	10%	WB	36.2	34.9	38.7	44.7	55.5	64.8	71.0	71.7	68.4	59.1	49.0	40.8
		MCDB	37.6	36.3	40.9	47.7	58.6	68.1	74.8	75.2	71.6	61.7	51.1	42.6
Mean Daily Temperature Range	5% DB	MDBR	13.0	12.0	11.1	10.9	11.4	11.5	11.6	12.1	12.6	10.7	10.4	11.2
		MCDBR	10.6	9.4	12.1	15.0	15.8	13.9	12.9	12.7	12.7	10.3	10.7	10.2
		MCWBR	10.7	8.9	9.6	10.3	10.5	9.0	8.7	8.6	9.0	8.8	9.7	10.1
	5% WB	MCDBR	10.3	9.3	11.6	13.3	14.1	12.0	11.6	11.2	11.4	9.8	10.4	9.8
		MCWBR	10.6	9.1	9.8	10.1	10.6	8.5	8.5	8.2	8.6	9.0	10.0	10.0
Clear-Sky Solar Irradiance	taub	0.298	0.305	0.321	0.363	0.387	0.427	0.427	0.416	0.375	0.352	0.311	0.288	
	taud	2.431	2.401	2.434	2.382	2.371	2.290	2.300	2.317	2.413	2.460	2.523	2.511	
	Ebn at Noon	264	282	292	286	281	269	267	266	269	262	259	259	
	Edh at Noon	23	29	32	36	38	41	40	38	32	27	22	20	
All-Sky Solar Radiation	RadAvg	405	661	1085	1430	1748	1858	1891	1666	1325	785	482	318	
	RadStd	42	61	96	156	172	149	149	88	108	70	61	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+133	N/A

Nomenclature: See separate page